Note

Deutrons (bundles of one proton and one neutron) can be accelerated because it has same charge as that of proton. But neutron (electrically neutral particle) cannot be accelerated by the cyclotron. When a deutron is bombarded with a beryllium target, a beam of high energy neutrons are produced. These high-energy neutrons are sent into the patient's cancerous region to break the bonds in the DNA of the cancer cells (killing the cells). This is used in treatment of fast-neutron cancer therapy.

EXAMPLE 3.23

Suppose a cyclotron is operated to accelerate protons with a magnetic field of strength 1 T. Calculate the frequency in which the electric field between two Dees could be reversed.

Solution

Magnetic field $B = 1$ T

Mass of the proton, $m_p = 1.67 \times 10^{-27}$ kg

Charge of the proton, $q = 1.60 \times 10^{-19}$ C

$$f = \frac{qB}{2\pi m_p} = \frac{\left(1.60 \times 10^{-19}\right)(1)}{2(3.14)(1.67 \times 10^{-27})}$$

$$= 15.3 \times 10^{6}\,\text{Hz} = 15.3\,\text{MHz}$$

3.10.5 Force on a current carrying conductor placed in a magnetic field

When a current carrying conductor is placed in a magnetic field, the force experienced by the conductor is equal to the sum of Lorentz forces on the individual charge carriers in the conductor. Consider a small segment of conductor of length *dl*, with cross-sectional area *A*

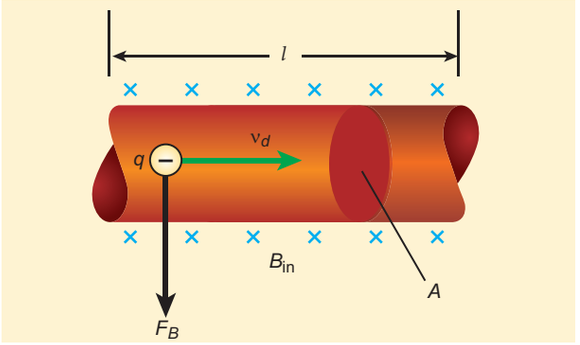


Figure 3.51 Current carrying conductor in a magnetic field

and current *I* as shown in Figure 3.51. The free electrons drift opposite to the direction of current. So the relation between current *I* and magnitude of drift velocity υ_d (Refer Unit 2) is

$$I = neA\upsilon_d \qquad (3.65)$$

If the conductor is kept in a magnetic field $\vec{B}$, then average force experienced by the charge (electron) in the conductor is

$$\vec{f} = -e\left(\vec{v}_d \times \vec{B}\right)$$

If *n* is the number of free electrons present in unit volume, then

$$n = \frac{N}{V}$$

where *N* is the number of free electrons in the small element of volume $V = Adl$.

Hence Lorentz force on the elementary section of length *dl* is the product of the number of the electrons ($N = nAdl$) and the force acting on each electron.

$$d\vec{F} = -enAdl\left(\vec{v}_d \times \vec{B}\right)$$

The current element in the conductor is $Id\vec{l} = -enA\vec{v}_d dl$. Therefore the force on the small elemental section of the current-carrying conductor is

$$d\vec{F} = \left(I\,d\vec{l} \times \vec{B}\right) \qquad (3.66)$$

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